Vegetation Cover Changes in the Khaeiz Protected Area Caused by Wildfire

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Class project: Remote Sensing and Landscape Planning (2020)

-1,200+ ha	1,415.52 ha	90 to 124	1.58 to 1.62
rangeland lost (27,853 to 26,204 ha)	classified as burned in 2020	rangeland patches (fragmentation)	rangeland SHAPE (more irregular)

Abstract

The Khaeiz Protected Area, part of Iran's Zagros mountains, hosts IUCN-listed and endangered species, including the Persian leopard, wild goat and Persian oak. A fire in June 2020 destroyed much of its vegetation. Using Landsat 8 images (2019 and 2020), land use maps and landscape metrics, this study quantifies the land changes caused by the fire. Rangeland fell from 27,853.11 ha (2019) to 26,203.77 ha (2020), a decrease of more than 1,200 ha. Bare mountains rose from 4,748.85 ha to 4,882.05 ha. The NP (number of patches) and PD (patch density) metrics show more agricultural and rangeland patches, indicating fragmentation of the rangeland by the fire.

1. Introduction

The Zagros forests are among Iran's most important forest habitats. Khaeiz is a hilly, mountainous, rocky area with 250-670 mm annual rainfall. It provides breeding habitat for rare wildlife (Persian leopard, wild goat/ibex, caracal, wolf, striped hyena, eagle owl), and its dominant plant is oak, with many oak taxa vulnerable or threatened (IUCN). The area holds more than 1,500 wild goats, an IUCN Endangered (EN) species; the Persian leopard is also EN.

The fire, probably human-caused, started on 8 Khordad 1399 and lasted 4 days. Satellite images and field visits showed the loss of more than 800 ha of natural land, rangeland and forest. Remote sensing is widely used to measure land cover change (Rogan et al., 2002; Seto et al., 2002). Azevedo et al. (2011) found that post-fire forest cover in a Portuguese mountain parish fell from 22% (1958) to under 5% (2005).

Aim: to examine vegetation cover change in Khaeiz between 2019 and 2020 using Landsat 8 satellite monitoring.

2. Materials and Methods

2.1 Study area

Hilly, mountainous and rocky, covering 32,332 ha, 15 km south of Behbahan. It is bounded to the east by Pol-e Padavak (Gachsaran) and to the west by the Maroun Dam. Elevation ranges from 460 to 1,500 m above sea level.



Figure 1. Location of the Khaeiz Protected Area in Iran.

2.2 Method

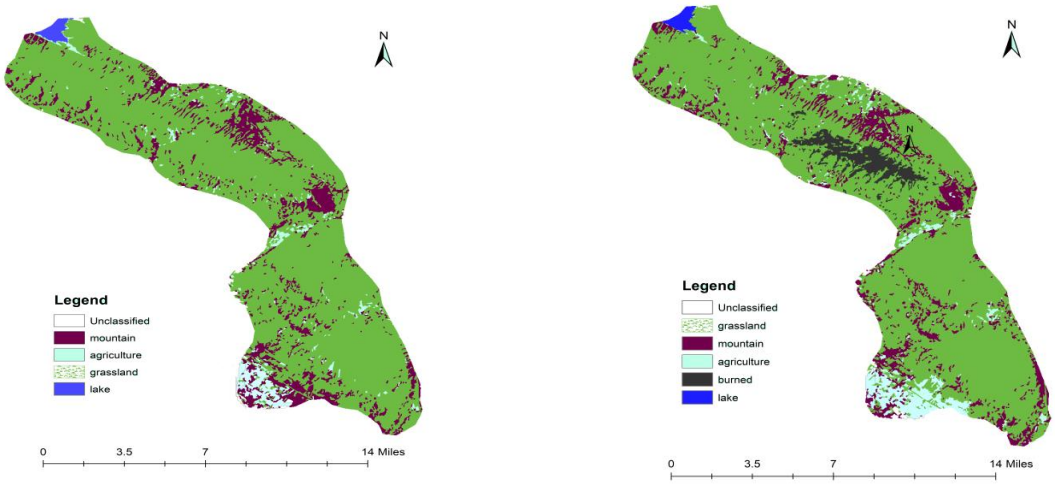
Landsat 8 (OLI and TIRS) images for 2019 and 2020. ENVI was used for classification, Fragstats for the metrics and GIS for mapping. FLAASH atmospheric correction was applied and accuracy was checked with an error matrix.

Landscape metrics

- **NP** (number of patches) = sum of patches of type i.
- **PD** (patch density) = NP / A x 10,000 x 100, where A = total landscape area in m^2.
- **SHAPE** = patch perimeter / minimum perimeter of a maximally compact shape of the same area. Values close to 1 mean a compact, regular shape; values above 1 mean a less compact, more irregular shape.

3. Results and Discussion

Total landscape area (TA, including the lake) is 34,279.02 ha in both years. The classes are bare mountains, agriculture and rural areas, rangeland and lake (2019), plus burned area (2020). Rangeland is the largest class and forms the landscape matrix; the lake is the smallest, at the northeast edge.



2019 *2020*

Figure 2. Classified maps of Khaeiz. Left: 2019. Right: 2020.

Table 1. Land use area by class

Class	2019 (ha)	2020 (ha)	Note
Rangeland	27,853.11	26,203.77	Decrease of more than 1,200 ha
Bare mountains	4,748.85	4,882.05	Increase
Agriculture and rural	1,427.22	1,532.34	Almost no change
Lake	247	247	Constant (average)
Burned area	-	1,415.52	Taken from rangeland

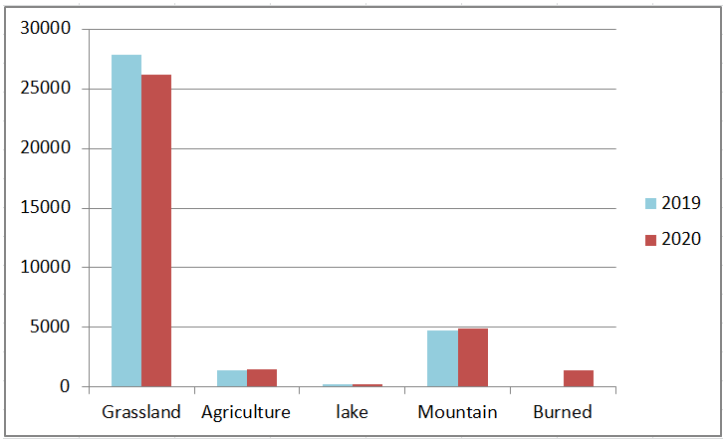


Figure 3. Land use change in Khaeiz, 2019 to 2020 (area in ha).

Table 2. Number of patches (NP) and patch density (PD)

Class	2019 NP	2019 PD	2020 NP	2020 PD
Rangeland	90	.26	124	.36
Agriculture	480	1.40	497	1.44
Lake	2	.005	57	.002
Bare mountains	581	1.69	571	1.66
Burned area	-	-	1	.16

Landscape effects

- **Rangeland patches:** NP rose from 90 to 124 and PD from .26 to .36, a small change in density. The rise in patch number probably reflects fragmentation of the rangeland by the fire.
- **Agriculture:** PD stayed constant but NP rose from 480 to 497, which may indicate more agricultural plots and expansion of agriculture in the area.
- **SHAPE:** rangeland rose from 1.58 to 1.62, the least compact and most irregular class and becoming more so. Agriculture went from 1.24 to 1.26, the most compact class.

Conclusion. The fire made the rangeland shape more irregular and increased its patch count, reducing its compactness and cohesion. Besides lowering the area's quantitative capacity, this change in ecosystem structure alters and disrupts ecosystem services, and recovery takes many years.

References

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